CSE 708
Graph Theory

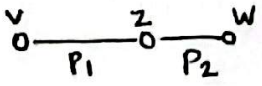
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$x = \text{mod}(21301002, 13) + 1 = 9$

5.9 (i) Let 'p' be the shortest path connecting vertex 'v' and 'w' in a connected graph. $\therefore$ length (p) = d(v,w)

since, $d(v,w) \geq 2$, the path p must contain two edges and three vertices.

Let select 'z' as the 'interal vertex.



here, P_1 is distance from v-z
P_2 is distance from z-w

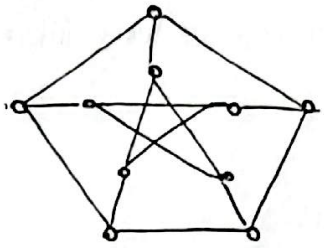
length $P_1 = d(v,z)$
length $P_2 = d(z,w)$

since the total length of a path is the sum of it split segments

length (p) = length (P_1) + length (P_2)

$\Rightarrow d(v,w) = d(v,z) + d(z,w)$

(ii)



* it has 10 vertices
* every vertex has 3 degree
* no cycles of length 3 or 4

lets pick an arbitrary vertex v and count how many vertices are reachable

1) Distance d = 0. This is just the starting vertex v itself
2) Distance d = 1. Vertex V is connected to exactly 3 distinct neighbors
3) Distance d = 2, Each of the 3 neighbors has 2 other edges leading new vertices. So we get, $3 \times 2 = 6$ edges branching forward.

As petersen graph has no cycles of length 3 or 4; none of those 6 paths can loop back to v, nor they can cross over and land on the same vertex.

Therefore, all 6 of those endpoints must lead to completely unique vertices.

Total Vertices reached = 1 (d=0) + 3 (d=1) + 6 (d=2) = 10 vertices

petersen graph only has 10 vertices, so every single vertex in the graph is accounted for within 2 distances from starting point.

Because V was choosen arbitrary, any two vertex v and w must either be directly adjacent $d(v,w) = 1$ or share a common neighbor $d(v,w) = 2$